

4) a) $P(X > Y)$

find the sum of the PMF over all pairs where $X > Y$. We use summation $1 \rightarrow 3$ (all y values) and an inside sum of $y+1 \rightarrow 5$ (all $x > y$ values).

$$P(X > Y) = \sum_{y=1}^3 \sum_{x=y+1}^5 \frac{1}{90} xy = \frac{1}{90} (1(2+3+4+5) + 2(3+4+5) + 3(4+5))$$

$$= \frac{1}{90} (65) = \boxed{\frac{65}{90}}$$

$$b) p_x(x) = \sum_{y=1}^3 \frac{1}{90} xy = \frac{x}{90} \sum_{y=1}^3 y = \frac{x}{90} (6) = \boxed{\frac{6x}{90}}$$

We just sum over y values, treating x as a constant.

$$c) p_y(y) = \sum_{x=1}^5 \frac{1}{90} xy = \frac{y}{90} \sum_{x=1}^5 x = \frac{y}{90} (15) = \boxed{\frac{15y}{90}}$$

d) They are independent iff $p_{xy}(x,y) = p_x(x)p_y(y)$

therefore: using answers from b and c,

$$\frac{15y}{90} \cdot \frac{6x}{90} = \frac{xy}{90} = p_{xy}(x,y)$$

They are independent.

$$c) E[\tilde{\theta}] = E\left[2 \sum_{i=1}^n Y_i\right]$$

linearity of expectation.

$$E[\tilde{\theta}] = \frac{2}{n} \sum_{i=1}^n E[Y_i]$$

$$Y_i \sim \text{Unif}(0, 1)$$

$$E[Y_i] = \frac{1}{2}$$

$$E[\tilde{\theta}] = \frac{2}{n} \sum_{i=1}^n \frac{1}{2} = \frac{2}{n} \cdot n \cdot \frac{1}{2} = 1$$

$$E[\tilde{\theta}] = 1$$

This is unbiased

$$3) a) P_{x,y}(x,y) = \begin{cases} 1/55 & 1 \leq x \leq y \leq 10, x, y \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

$$a) P(y|x) = \sum_x P_{x,y}(x,y)$$

From the PMF, we know that y ranges from x to 10.

$$P_x(x) = \sum_x P_{x,y}(x,y) = \sum_x \frac{1}{55}$$

$$= \left[\frac{1}{55} \cdot (10 - x + 1) \right]$$

$10 - x + 1$ is the total number of terms from $x \rightarrow 10$.

$$P_x(x) = \boxed{\frac{11-x}{55}}$$

$$P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)} = \frac{1/55}{(11-x/1/55)} = \boxed{\frac{1}{11-x}}$$

plug into formula:

$$\begin{aligned} \text{b) } E[Y|X=x] &= \sum_x^{10} y \cdot P_{Y|X}(y|x) = \sum_x^{10} y \cdot \frac{1}{11-x} \\ &= \frac{1}{11-x} \sum_x^{10} y \quad \sum_x^{10} y = \frac{(11-x)(x+10)}{2} \\ &\text{substitute:} \end{aligned}$$

$$= \left(\frac{1}{11-x} \right) \cdot \left(\frac{(11-x)(x+10)}{2} \right)$$

$$E[Y|X=x] = \boxed{\frac{x+10}{2}}$$

$$\begin{aligned} \text{c) } E[Y] &= \sum_y P_Y(y) \\ P_Y(y) &= \sum_x P_{X,Y}(x,y) \\ &= \sum_x^{10} \frac{1}{55} = \frac{y}{55} \quad \text{for } y \in \{1, 2, \dots, 10\} \\ E[Y] &= \sum_y^{10} y \cdot P_Y(y) = \sum_y^{10} y \cdot \frac{y}{55} \\ &= \frac{1}{55} \sum_y^{10} y^2 = \text{formula for sum of squares} \end{aligned}$$

$$= \frac{1}{55} \left(\frac{10(11)(21)}{6} \right) = \boxed{\frac{385}{55}}$$

$$d) \quad \hat{\theta}_H = \frac{x}{n}$$

$$\hat{\theta}_D = \frac{y}{n}$$

2 Unbiased Estimators

$$a) \quad L(y_1, \dots, y_n; \theta) \sim \text{Unif}(0, \theta)$$

$$= L(y_n; \theta) = \prod_{i=1}^n f(y_i; \theta) \quad f(y_i; \theta) = \begin{cases} \frac{1}{\theta} & 0 < y_i < \theta \\ 0 & \text{otherwise} \end{cases}$$

$$= \prod_{i=1}^n \frac{1}{\theta} = \frac{1}{\theta^n}$$

Since $\frac{1}{\theta^n}$ is maximized when θ^n is as small as possible, we know that the largest $(y_1, \dots, y_n)$ value is where the function will be maximized. We can't set the derivative to 0, because this function has no "peak" as it only can decrease for $y > 0$.

$$\hat{\theta} = \max(y_1, y_2, \dots, y_n)$$

b) Biased: $E[\hat{\theta}] = 0$

$$\hat{\theta} = \max(y_n) \quad E[\hat{\theta}] =$$

First we find the cdf of $\hat{\theta}$ so we can find the pdf.

CDF: $(F_n(y))^n$ (following the section 7 logic).

$$(F_n(y))^n = \left(\frac{y}{\theta}\right)^n$$

PDF: Take derivative

$$\frac{d}{dy} \left(\frac{y}{\theta} \right)^n = \frac{n y^{n-1}}{\theta^n}$$

$$E[\hat{\theta}] = \int_0^{\theta} y \cdot \frac{n y^{n-1}}{\theta^n} dy$$

$$= \frac{n}{\theta^n} \int_0^{\theta} y^n dy = \frac{n}{\theta^n} \left[\frac{y^{n+1}}{n+1} \right]_0^{\theta} = \frac{n}{\theta^n} \left(\frac{\theta^{n+1}}{n+1} \right) = \left(\frac{n}{n+1} \right) \theta$$

$$E[\hat{\theta}] = \left(\frac{n}{n+1} \right) \theta$$

Since $\frac{n}{n+1}$ isn't 1, $E[\hat{\theta}] \neq \theta$. Therefore it is

biased.

1) a)

$$L(y_1, \dots, y_n; \theta_A, \theta_B) = \prod_1^n P(y_i; \theta)$$

$$= \theta_A^x \cdot \theta_B^y \cdot (1 - \theta_A - \theta_B)^z$$

We multiply the probabilities together, we have exponents x, y, z since we have x instances of Alice, y instances of Bob, and z instances of Calvin.

$$b) \ln(L(y_1, \dots, y_n; \theta_A, \theta_B)) = \ln(\theta_A^x \cdot \theta_B^y \cdot (1 - \theta_A - \theta_B)^z)$$

$$= x \ln(\theta_A) + y \ln(\theta_B) + z \ln(1 - \theta_A - \theta_B)$$

c) take natural log, and apply property that $\ln(ab) = \ln a + \ln b$

c) Take Partial derivatives with respect to θ_A and θ_B

$$\frac{\partial}{\partial \theta_A} : \frac{x}{\theta_A} + 0 + \frac{z}{(1 - \theta_A - \theta_B)} (-1) = 0$$

$$\frac{\partial}{\partial \theta_B} : 0 + \frac{y}{\theta_B} + \frac{z}{(1 - \theta_A - \theta_B)} (-1) = 0$$

$$\frac{x}{\theta_A (1 - \theta_A - \theta_B)} = \frac{y}{\theta_B} \quad \text{Therefore } \frac{x}{\theta_A} = \frac{y}{\theta_B}$$

$$\theta_B = \theta_A \frac{y}{x}$$

$$\theta_A = \theta_B \frac{x}{y}$$

$$c) E[\tilde{\theta}] = E\left[\frac{2}{n} \sum_{i=1}^n Y_i\right]$$

linearity of expectation.

$$E[\tilde{\theta}] = \frac{2}{n} \sum_{i=1}^n E[Y_i]$$

$$Y_i \sim \text{Unif}(0, \theta)$$

$$E[Y_i] = \frac{\theta}{2}$$

$$E[\tilde{\theta}] = \frac{2}{n} \sum_{i=1}^n \frac{\theta}{2} = \frac{2}{n} \cdot n \cdot \frac{\theta}{2} = \theta$$

$$E[\tilde{\theta}] = \theta$$

This is unbiased

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$$= \frac{1}{55} \cdot (10 - x + 1)$$

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$$P_x(x) = \frac{11-x}{55}$$